

Towards a Linear-Algebraic Hypervisor

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The RASP Model

Definition (Cook & Reckhow, 1973)

A RASP machine, $\mathcal{R} = \langle C, c_0, \rightarrow, F \rangle$ consists of a configuration space, $\langle i, a, M, u, y \rangle : C = \mathbb{N} \times \mathbb{Z} \times \mathbb{Z}^{\mathbb{N}} \times \mathbb{Z}^* \times \mathbb{Z}^*$. A RAS program is a vector $P : (\mathbb{Z} \times \mathbb{Z})^m$ packed with an *input* $x : \mathbb{Z}^*$, into an *initial configuration*, $c_0 = \langle 0, 0, M'_{0..2m} \leftarrow P, x0, \varepsilon \rangle$. Let $\langle o, j \rangle = \langle M_i, M_{i+1} \rangle$ denote the *opcode* and *operand* and $\rightarrow$ be defined as:

$$\langle i, a, M, u, y \rangle \rightarrow \begin{cases} \langle i+2, j, M, u, y \rangle, & \text{if } o = 1_{\text{LOD}}, \\ \langle i+2, a + M_j, M, u, y \rangle, & \text{if } o = 2_{\text{ADD}}, \\ \langle i+2, a - M_j, M, u, y \rangle, & \text{if } o = 3_{\text{SUB}}, \\ \langle i+2, a, M_j \leftarrow a, u, y \rangle, & \text{if } o = 4_{\text{STO}}, \\ \langle j, a, M, u, y \rangle, & \text{if } o = 5_{\text{BPA}} \wedge a > 0, \\ \langle i+2, a, M, u, y \rangle, & \text{if } o = 5_{\text{BPA}} \wedge a \leq 0, \\ \langle i+2, a, M_j \leftarrow \sigma, v, y \rangle, & \text{if } o = 6_{\text{RD}} \wedge u = \sigma v, \\ \langle i+2, a, M, u, y \cdot M_j \rangle, & \text{if } o = 7_{\text{PRI}}, \\ \langle i, a, M, u, y \rangle, & \text{otherwise.} \end{cases}$$

and the halting configurations are simply $F = \{c \in C \mid c \rightarrow c\}$.

W-RASP: A Bounded Variant

Fix $w, m, n, \ell, s \in \mathbb{N}$, assume $\mathbb{W} = \mathbb{F}_2^w$, and for $P : \mathbb{W}^{2m \leq n}$ define:

$$\mathcal{R}^{(w,n,\ell,s)} = \langle C_w, c_0, \rightarrow, F \rangle, \quad C_w = \mathbb{W} \times \mathbb{W} \times \mathbb{W}^n \times \mathbb{W}^{\ell+1} \times \mathbb{W}^{s+1}.$$

The transition operator takes a configuration $\langle i, a, M, u, y \rangle : C_w$ to:

$$\left\{ \begin{array}{ll} \langle i \oplus 2_w, j, M, u, y \rangle, & \text{if } o = 1_{\text{LOD}}, \\ \langle i \oplus 2_w, a \oplus M_{j \bmod n}, M, u, y \rangle, & \text{if } o = 2_{\text{ADD}}, \\ \langle i \oplus 2_w, a \otimes M_{j \bmod n}, M, u, y \rangle, & \text{if } o = 3_{\text{MUL}}, \\ \langle i \oplus 2_w, a, M_{j \bmod n} \leftarrow a, u, y \rangle, & \text{if } o = 4_{\text{STO}}, \\ \langle j, a, M, u, y \rangle, & \text{if } o = 5_{\text{BNZ}} \wedge a \neq 0, \\ \langle i \oplus 2_w, a, M, u, y \rangle, & \text{if } o = 5_{\text{BNZ}} \wedge a = 0, \\ \langle i \oplus 2_w, a, M_{j \bmod n} \leftarrow u^\bullet, u^\blacktriangleright, y \rangle, & \text{if } o = 6_{\text{RD}} \wedge u_0 < \ell, \\ \langle i \oplus 2_w, a, M, u, y :: M_{j \bmod n} \rangle, & \text{if } o = 7_{\text{PRI}}, \\ \langle i, a, M, u, y \rangle, & \text{otherwise.} \end{array} \right.$$

$$\langle o, j \rangle := \langle M_{i \bmod n}, M_{(i \oplus 1) \bmod n} \rangle, \quad M_j \leftarrow a := \left[\delta_{jk} a + (1 - \delta_{jk}) M_k \right]_{k=0}^{n-1}.$$

$$u^\bullet := u_{\min(u_0+1, \ell)}, \quad u^\blacktriangleright := u_0 \leftarrow \min(u_0 + 1, \ell),$$

$$y :: z = \begin{cases} (y_{y_0+1} \leftarrow z)_0 \leftarrow y_0 + 1, & y_0 < s, \\ y, & \text{otherwise.} \end{cases}$$

Heapless Array Language

```
F ::= fun f0 ( ipt : W ^ N ) -> W ^ N { B }
N ::= 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9
B ::= S | S ; B | hlt | brk
S ::= P = E | ife G { B } { B } | whl G { B }
Z ::= 0 | N
G ::= ipt [ Z ] | scr [ Z ] | opt [ Z ]
P ::= scr [ Z ] | opt [ Z ]
E ::= G + G | G * G | G + Z | G * Z | G | Z
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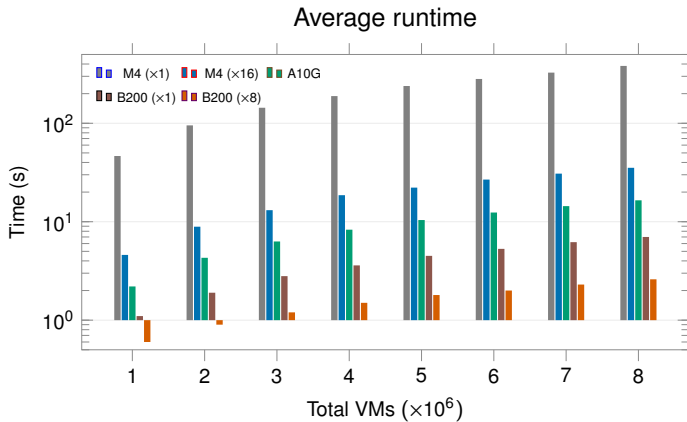
Reserve disjoint memory regions $p_{0\dots L-1}$, $q_{0\dots e-1}$, and $r_{0\dots \gamma-1}$ for arrays $\text{ipt} : \mathbb{W}^{L \leq \ell}$, $\text{opt} : \mathbb{W}^{e \leq s}$ and $\text{scr} : \mathbb{W}^{\gamma \leq \mu}$, with $(5 + \ell + s + \mu + 2m) \leq n$ and interpret array indexing modulo declared length, so $\llbracket \alpha \rrbracket_z := M_{2m + \llbracket \alpha \rrbracket + (z \bmod |\alpha|)}$.

Heapless Array Lowering

$$\begin{array}{c}
 \frac{\Gamma[k \mapsto \ell_f] \vdash B \Rightarrow C}{\vdash \text{fun } f0 \text{ (} \text{ipt} : I \wedge N \text{) } \rightarrow I \wedge L \{ B \} \Rightarrow C} \text{FUN} \quad \ell_f: \boxed{\text{PRI } q_i}_{i=0}^{l-1} \boxed{\text{HLT}} \\
 \\
 \frac{}{\Gamma \vdash \alpha[z] \Rightarrow_a \boxed{\text{LOD } 0} \boxed{\text{ADD } \llbracket \alpha \rrbracket_z}} \text{GET} \quad \frac{\Gamma \vdash E \Rightarrow_a A}{\Gamma \vdash \alpha[z] = E \Rightarrow A \boxed{\text{STO } \llbracket \alpha \rrbracket_z}} \text{PUT} \\
 \\
 \frac{\Gamma \vdash l_1 \Rightarrow_a J_1 \quad \Gamma \vdash l_2 \Rightarrow_a J_2}{\Gamma \vdash l_1 + l_2 \Rightarrow_a J_1 \boxed{\text{STO } \nu} J_2 \boxed{\text{ADD } \nu}} \oplus \quad \frac{\Gamma \vdash l_1 \Rightarrow_a J_1 \quad \Gamma \vdash l_2 \Rightarrow_a J_2}{\Gamma \vdash l_1 \times l_2 \Rightarrow_a J_1 \boxed{\text{STO } \nu} J_2 \boxed{\text{MUL } \nu}} \otimes \\
 \\
 \frac{\Gamma \vdash l \Rightarrow_a J \quad \Gamma \vdash B_1 \Rightarrow C_1 \quad \Gamma \vdash B_2 \Rightarrow C_2}{\Gamma \vdash \text{ife } l \{B_1\} \{B_2\} \Rightarrow J \boxed{\text{BNZ } \ell_t} C_2 \boxed{\text{JMP } \ell_e} \ell_t: C_1 \ell_e:} \text{IFE} \\
 \\
 \frac{\Gamma \vdash l \Rightarrow_a J \quad \Gamma[k \mapsto \ell_e] \vdash B \Rightarrow C}{\Gamma \vdash \text{whl } l \{B\} \Rightarrow \ell_h: J \boxed{\text{BNZ } \ell_b} \boxed{\text{JMP } \ell_e} \ell_b: C \boxed{\text{JMP } \ell_h} \ell_e:} \text{WHILE} \quad \frac{\Gamma(k) = \ell}{\Gamma \vdash \text{brk} \Rightarrow \boxed{\text{JMP } \ell;}} \text{BREAK} \\
 \\
 \frac{\Gamma \vdash B \Rightarrow C_B \quad \Gamma \vdash S \Rightarrow C_S}{\Gamma \vdash B ; S \Rightarrow C_B C_S} \text{SEQ} \quad \frac{}{\Gamma \vdash \text{hlt} \Rightarrow \boxed{\text{PRI } q_i}_{i=0}^{l-1} \boxed{\text{HLT}}} \text{HALT}
 \end{array}$$

We write $\Gamma \vdash E \Rightarrow A$ when the A implements E , and $\Gamma \vdash E \Rightarrow_a A$ when A implements E , leaving value a in the accumulator. Let ν be a reserved memory address and desugar $\boxed{\text{JMP } \ell}$ as $\boxed{\text{LOD } 1} \boxed{\text{BNZ } \ell}$.

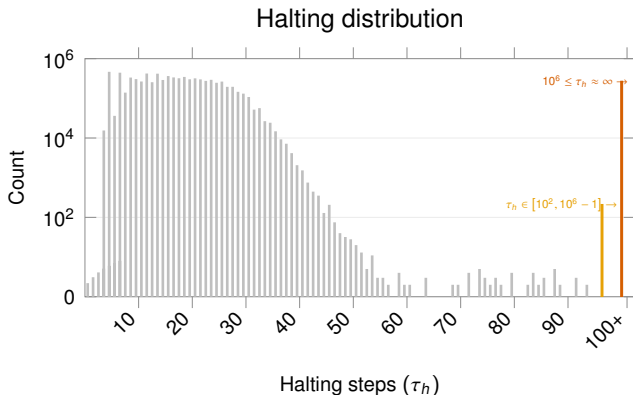
Evaluation: Runtime



$w = 32, n = 250, \ell = 10, s = 2, \tau_{\max} = 10^6, d = 8 \times 10^6, L = 100.$

Evaluation: Halting Distribution

$$\Pr_{P' \sim \mathcal{P}_L} \left(\Phi_w^{\tau_{\max}}(c_0(P' \Rightarrow P, x \sim \mathbb{F}_2^{w \cdot \text{arity}(P')})) \in F \mid L = |P'| \right).$$



We plot programs by their halting number τ_h , the least τ such that $\Phi_w^\tau(c_0) = \Phi_w^{\tau+1}(c_0)$.